CS342 Coursework

Predicting Battery Useful Life

```
# setup and libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# import data
battery_metadata = pd.read_csv('battery_metadata.csv')
battery_observations = pd.read_csv('battery_observations.csv')
training_data = pd.read_csv('train_batteries.csv')
validation_data = pd.read_csv('validation_batteries.csv')
test_data = pd.read_csv('test_batteries.csv')
```

Task 1: Exploratory Data Analysis and Data Visualisation

Task 2: Unsupervised (Classic) Machine Learning for Battery Degradation Analysis

Task 3: Supervised (Classic) Machine Learning for Battery Degradation Analysis

Task 4: Unsupervised Deep Learning for Battery Degradation Analysis

Task 5: Supervised Deep Learning for Battery Degradation Analysis